

Automatic Transport Selection Ladder

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Master technical specification — Volume 2 (Data Plane), nano-detail deep-dive. This document **deepens** the *Auto transport selection — the escalation ladder* section of doc 01 [01-DP §5.3, §5.4] into an implementation-ready specification of the ladder state machine, its timers, per-network memory, coordinator-pushed regional priors, manual override, telemetry, and the byte layouts of every persisted/streamed artefact. SPEC-ONLY: it describes **what to build**, not the shipping product. Sources cited inline by id — [01-DP §N] = doc 01 final/01-data-plane.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_P0]/[04_P2] = Phase-0/Phase-2 refined docs; [research-masque], [research-hysteria2], [research-daita_test] = the cited research digests; [SYNTHESIS §N] = the cross-document synthesis. Any claim not grounded in the evidence base is marked UNVERIFIED per constitution §11.4.6.

0. Position, ownership, and invariants

0.1 What this document owns

The ladder is the **client-side control logic** that, given an ordered list of candidate L2 transports (plain-udp → lwo → masque-h3 → shadowsocks → udp-over-tcp, with hysteria2 and connect-ip as feature-gated rungs), picks the one that actually carries WireGuard datagrams to the gateway on the current network, and

re-picks automatically when a rung stops working [01-DP §5.3, §3.1]. It is the code behind `helix-core/src/ladder.rs` [01-DP §12]. It owns:

1. The **ladder state machine** + its transition timers/budgets (§3, §4).
2. The driver that calls `helix_transport::dial()` and reacts to `TransportError` (§5).
3. **Per-network memory** — remember-what-worked keyed by network fingerprint (§6).
4. Consumption of the **coordinator-pushed regional prior** `TransportPolicy.order` (§7).
5. **Manual override** (pin) — Mullvad's "force this transport" toggle (§8).
6. **Aggregate, no-per-user telemetry** of escalation outcomes (§9, invariant I5).

0.2 What this document does NOT own

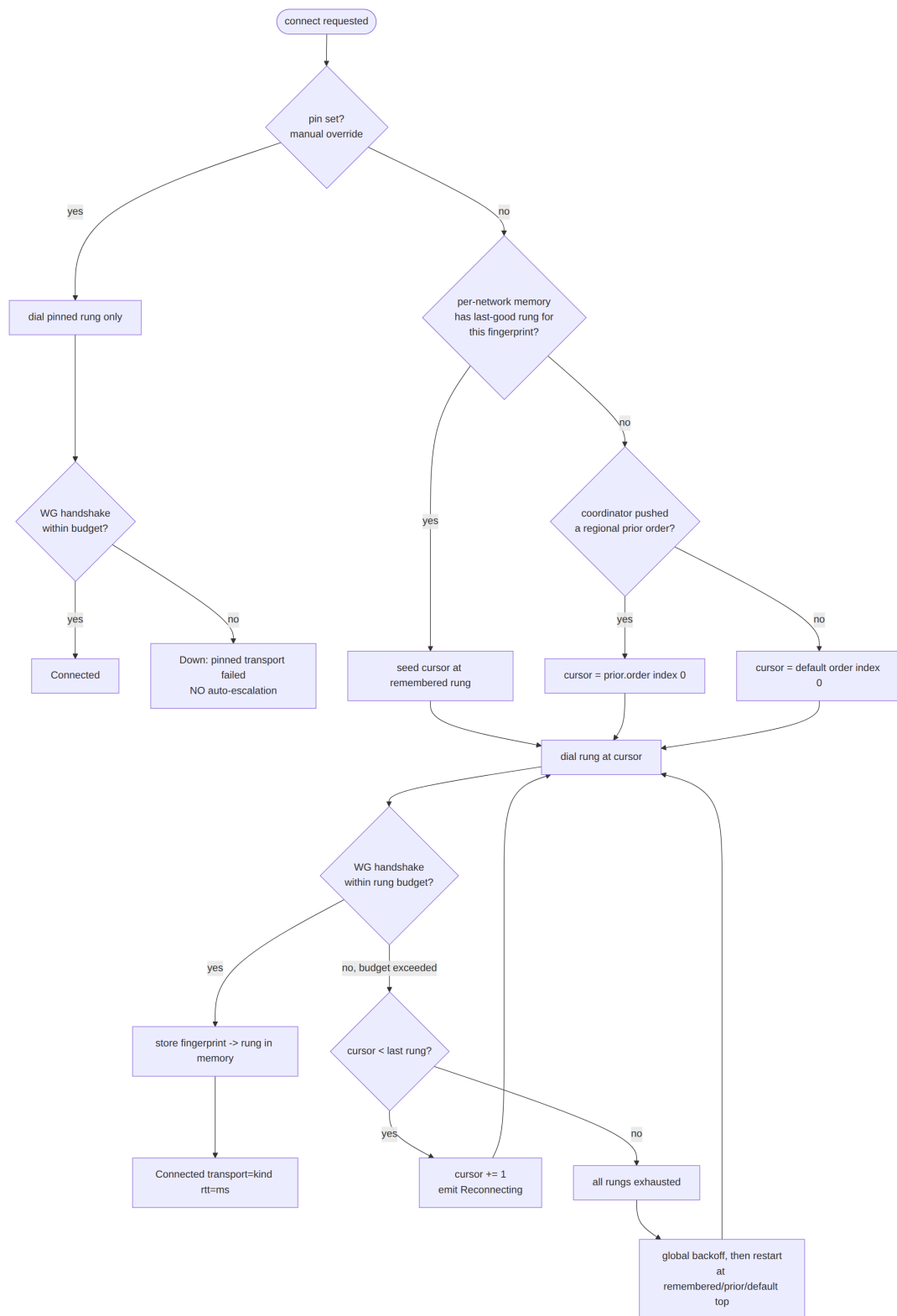
It does **not** define the `Transport` trait, `TransportConfig`, `dial()`, or the wire framing of any transport — those are frozen contracts in doc 01 [01-DP §3]. It does **not** own the `WatchNetworkMap` protobuf that *delivers* the regional prior — that is doc 03; this document specifies only the **shape the ladder consumes** and treats the protobuf as the upstream source (§7.1). It does **not** own WireGuard crypto (`helix-wg` [01-DP §4]), routing/ACL (`helix-route` [01-DP §6–§7]), DAITA (`helix-daita` [01-DP §9]), or the FFI surface (doc 05). It surfaces `TunnelStatus` events but does not own the FFI bridge that re-emits them.

0.3 Invariants this document inherits and tightens

#	Invariant	Source	Ladder-specific tightening
I1	Transport never sees plaintext.	[01-DP I1]	The ladder operates on opaque <code>Box<dyn Transport></code> handles + health snapshots; it never inspects datagram bytes.
I2	Unreliable datagrams, never an ordered stream.	[01-DP I2]	The ladder's success criterion is a completed WG handshake over the rung, not a TCP/QUIC connect — a connected carrier with no WG

#	Invariant	Source	Ladder-specific tightening
			handshake is not a successful rung (§5.5).
I5	No-logging by construction; only aggregate counters.	[01-DP I5, 04_ARCH §2.7]	Per-network memory persists a fingerprint hash + last-good rung index , never the SSID plaintext, never a server endpoint, never a timestamped per-connection log (§6.4, §9).
L-17	Automatic with manual override — Mullvad's exact UX.	[04_ARCH §3.2, 01-DP §5.3]	pin short-circuits the ladder entirely; auto mode walks order with per-rung budgets (§3, §8).
L-18	Escalation is monotonic within one connect attempt — never oscillate up-and-down inside a single attempt; only reset to the top on a new attempt or successful reconnect.	derived from [04_P2 §1.4]	Prevents thrash; §3.3, §4.4.
L-19	Every PASS carries captured evidence (the escalation event trace IS the evidence).	[01-DP §12.1, §11.4.5/.69/.107]	§9, §10.

1. The ladder model in one picture



The cursor only ever **advances** within one attempt (L-I8). The three seed sources — per-network memory, regional prior, default — are tried in that priority order (§2.3). pin bypasses all of it (§8).

2. Ordered rung taxonomy & the policy object

2.1 TransportKind — the ordinal the ladder walks

```
// helix-core/src/ladder.rs
// The closed set of ladder rungs. Ordinal order is the *default*
// escalation order
// (cheapest/fastest first → most evasive/most expensive last)
// [01-DP §5.3, 04_P2 §1.4].
#[derive(Clone, Copy, Debug, PartialEq, Eq, Hash, PartialOrd,
         Ord)]
#[repr(u8)]
pub enum TransportKind {
    PlainUdp    = 0, // baseline, lowest latency/CPU [01-DP
                    §3.2]
    Lwo         = 1, // cheap header obfuscation, first
                    escalation rung [01-DP §3.7]
    MasqueH3    = 2, // CONNECT-UDP / HTTP-3 – primary
                    obfuscation (D1 camp A) [01-DP §3.3]
    Hysteria2   = 3, // QUIC+Salamander/Gecko,
                    feature="hysteria2" (D1 camp B) [01-DP §3.8]
    Shadowsocks = 4, // WG-in-Shadowsocks AEAD over TCP [01-DP
                    §3.5]
    UdpOverTcp  = 5, // last resort when ALL udp/quic blocked
                    [01-DP §3.6]
    ConnectIp   = 6, // RFC 9484 IP-over-H3, feature="connect-
                    ip", advanced [01-DP §3.4]
}

impl TransportKind {
    /// Stable wire/metric label – MUST equal the matching
    /// `Transport::kind()` string [01-DP §3].
    pub const fn label(self) -> &'static str {
        match self {
            TransportKind::PlainUdp => "plain-udp",
            TransportKind::Lwo      => "lwo",
            TransportKind::MasqueH3 => "masque-h3",
            TransportKind::Hysteria2 => "hysteria2",
            TransportKind::Shadowsocks => "shadowsocks",
            TransportKind::UdpOverTcp => "udp-over-tcp",
            TransportKind::ConnectIp => "connect-ip",
        }
    }

    /// Compact persistence/telemetry code (1 byte) – the
    /// discriminant.
    pub const fn code(self) -> u8 { self as u8 }
    pub fn from_code(b: u8) -> Option<TransportKind> { /* match
        0..=6 → variant; else None */ }

    /// Carrier requires UDP/QUIC reachability (fails on a hard
    /// UDP block) – used by the
```

```

    /// network-class hint (§6.5) to skip provably-dead rungs
    /// early.
    /// plain-udp/lwo/masque-h3/hysteria2/connect-ip = true;
    /// shadowsocks/udp-over-tcp = false
    /// [research-hysteria2 §5(b): "QUIC/Hysteria2/MASQUE-over-H3/
    /// WireGuard all fail ... need UDP"].
    pub const fn needs_udp(self) -> bool {
        !matches!(self, TransportKind::Shadowsocks |
        TransportKind::UdpOverTcp)
    }
}

```

Why this ordinal order (each rung is strictly more evasive *and* more expensive than the prior): plain-udp is the $\geq 80\%$ -bare-link baseline [01-DP §3.2, 04_P0 §0]; lwo is "near-zero-cost evasion of naive WG fingerprinting" [01-DP §3.7]; masque-h3 "genuinely is HTTP/3 to an HTTP proxy" (strong blend-in) but pays the double-crypto/double-CC throughput tax [research-masque §2; research-hysteria2 §6]; hysteria2 wins raw throughput on lossy/throttled links (Brutal CC) but is a non-standard "random noise" wire [research-hysteria2 §1, §6]; the two TCP carriers carry the head-of-line-blocking penalty and exist purely to keep a tunnel *possible* [01-DP §3.5, §3.6, 04_P2 §1.2]. connect-ip is last and feature-gated because it loses I1 at the gateway [01-DP §3.4].

DECISION D1 dependency. Where hysteria2 sits in the default order is contingent on D1 [01-DP §3.8, SYNTHESIS §3 D1]. This spec places it *after* masque-h3 (adopting Camp A: masque-h3 primary) and only when feature = "hysteria2" is compiled in; if the project later resolves D1 to Camp B, the coordinator's regional prior (§7) re-orders it to index 0 for the relevant regions **without a client rebuild** — the ladder reads order from policy, not from the enum ordinal (§2.2). The enum ordinal is only the *fallback default* when no prior is pushed.

2.2 TransportPolicy — the per-leg ladder configuration

This refines the doc-01 sketch [01-DP §5.3] with the fields the ladder actually needs:

```

// helix-core/src/ladder.rs
#[derive(Clone, Debug)]
pub struct TransportPolicy {
    /// The escalation order to walk in auto mode. Source priority
    /// (§2.3):
    ///   coordinator regional prior > compiled default. Never
    ///   empty (validated, §2.4).
    pub order: Vec<TransportKind>,

    /// Manual override (Mullvad "force transport X"). When
    /// `Some`, the ladder dials ONLY

```

```

    /// this rung and NEVER auto-escalates (§8). Survives
    reconnects until the user clears it.
    pub pin: Option<TransportKind>,

    /// Per-rung failure budget before escalating to the next rung
    (§4).
    pub budget: FailureBudget,

    /// Global backoff applied after the WHOLE order is exhausted,
    before restarting (§4.3).
    pub exhaust_backoff: Backoff,

    /// Monotonic version of the policy (from the regional-prior
    delta, §7.1). The ladder
    /// adopts a higher-versioned policy mid-session on the next
    attempt boundary, never
    /// mid-attempt (L-I8). Lower/equal version deltas are ignored
    (§7.4).
    pub version: u64,

    /// Provenance for telemetry + UX ("why is my order what it
    is"). No per-user data.
    pub source: PolicySource,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub enum PolicySource { CompiledDefault, RegionalPrior,
    OperatorOverride }

impl TransportPolicy {
    /// The compiled fallback used when no prior has arrived yet
    (e.g. first launch, offline).
    /// Unrestricted-network default: try the fast path first,
    escalate on failure [01-DP §5.3].
    pub fn compiled_default() -> Self {
        TransportPolicy {
            order: vec![
                TransportKind::PlainUdp,
                TransportKind::Lwo,
                TransportKind::MasqueH3,
                #[cfg(feature = "hysteria2")]
                TransportKind::Hysteria2,
                TransportKind::Shadowsocks,
                TransportKind::UdpOverTcp,
            ],
            pin: None,
            budget: FailureBudget::default(),
            exhaust_backoff: Backoff::default(),
            version: 0,
            source: PolicySource::CompiledDefault,
        }
    }
}

```

2.3 Seed-source priority (which rung the ladder STARTS at)

For a given connect attempt the **starting cursor** is resolved deterministically (no guessing, §11.4.6):

1. if policy.pin.is_some() → dial pin ONLY, no ladder (§8)
2. else if per-network memory hit → start cursor at remembered rung (§6) [01-DP §5.3 step 4]
(clamped into policy.order; if remembered rung absent from order → fall through)
3. else if policy.source == RegionalPrior → start cursor at order[0] (§7) [01-DP §5.3 step 5]
4. else → start cursor at order[0] (default) (§2.2)

Per-network memory **out-ranks** the regional prior because a remembered success on *this exact network* is stronger evidence than a region-wide statistical prior — the difference between "VPN that sometimes connects" and "VPN that just works" [01-DP §5.3 step 4, 04_P2 §1.4]. The regional prior still governs the *order of escalation* if the remembered rung subsequently fails (§6.3).

2.4 Policy validation (fail-loud, never silent)

order MUST be non-empty, contain no duplicate TransportKind, and reference only rungs the build actually compiled (e.g. Hysteria2/ConnectIp only when their feature is enabled). A regional prior that references an uncompiled rung is **filtered** (the rung is dropped from order with a counted policy_rung_unavailable telemetry tick, §9), not rejected wholesale — so a partial-feature client still gets a usable subset. An *empty-after-filter* order falls back to compiled_default() (§2.2) and counts policy_empty_fallback. Validation runs on every adopted policy delta (§7.4).

3. The ladder state machine

3.1 States

```
// helix-core/src/ladder.rs
#[derive(Clone, Debug)]
pub enum LadderState {
    /// No attempt in flight; awaiting a `start` command. Cursor
    /// undefined.
    Idle,
    /// Resolving the seed source (§2.3) and constructing the next
    /// TransportConfig.
```



```

Selecting { cursor: usize, attempt_id: u64 },
/// `dial()` in flight for `order[cursor]`; waiting up to
    `budget.dial_timeout`.
Dialing { cursor: usize, attempt_id: u64, rung_started:
    Instant, dial_tries: u8 },
/// Carrier up; waiting for the WG handshake to complete over
    it (§5.5),
/// up to `budget.handshake_timeout`.
Handshaking { cursor: usize, attempt_id: u64, rung_started:
    Instant },
/// WG handshake completed → tunnel live. Steady state. Holds
    the live transport.
Connected { cursor: usize, kind: TransportKind, rtt_ms: u32,
    since: Instant },
/// A rung failed within budget; escalating to
    `order[cursor+1]` on the same attempt (L-I8).
Escalating { from: usize, to: usize, attempt_id: u64, reason:
    EscalationReason },
/// Entire order exhausted; in `exhaust_backoff` before a
    fresh attempt (new attempt_id).
Exhausted { attempt_id: u64, until: Instant },
/// Connected tunnel lost liveness (no recv past
    `liveness_grace`); re-running the ladder
/// from the seed source while keeping the old transport for
    fail-static forwarding (§5.6).
Reconnecting { prev_kind: TransportKind, attempt_id: u64 },
/// Terminal-for-now: pinned transport failed (no escalation,
    §8) OR a non-retryable error.
Down { reason: DownReason },
}

```

attempt_id is a monotonic u64 minted at each *fresh* top-of-ladder entry (start, exhaust-restart, reconnect). It scopes per-attempt budgets and lets late dial() results from a superseded attempt be discarded (§5.4 cancel-safety).

3.2 LadderState → TunnelStatus projection

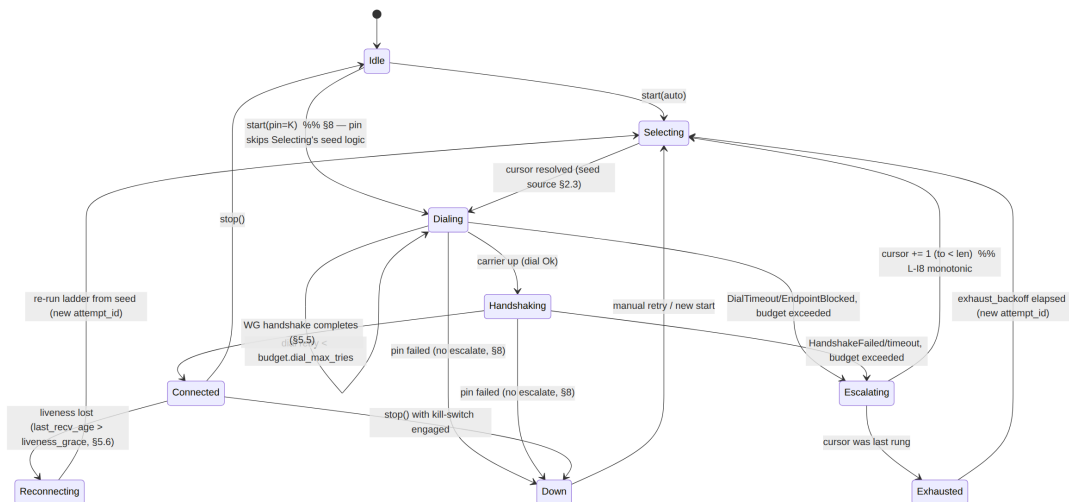
The ladder never owns the FFI; it projects its internal state onto the doc-01 TunnelStatus broadcast enum [01-DP §5.2] so Flutter/metrics/telemetry see a stable surface:

LadderState	Emitted TunnelStatus
Idle	(none / Down{reason:"idle"} on first stop)
Selecting, Dialing	Connecting
Handshaking	Handshaking
Connected{kind,rtt_ms,...}	Connected { transport: kind.label().into(), rtt_ms }
Escalating, Exhausted	Reconnecting
Reconnecting	Reconnecting

LadderState	Emitted TunnelStatus
Down{reason}	Down { reason: reason.to_string() }

The **ordered sequence** of these events on one escalation is itself the captured-evidence trace the acceptance gate checks (§9, §10.4) [01-DP §12.1].

3.3 State diagram



Connected → Reconnecting is the **only** downgrade path, and it always re-enters at the seed source (memory/prior/default), never mid-cursor — preserving L-I8 across the connected→lost boundary.

4. Timers, budgets, and backoff

4.1 FailureBudget — per-rung failure tolerance

A rung is abandoned when it exhausts its budget; the budget is **per rung, per attempt** [01-DP §5.3, 04_P2 §1.4]. The doc-01 sketch { max_handshakes: u8, window: Duration } [01-DP §5.3] is refined to separate the *dial* phase (is the carrier reachable) from the *handshake* phase (does WG complete):

```

// helix-core/src/ladder.rs
#[derive(Clone, Copy, Debug)]
pub struct FailureBudget {
    /// Max wall-clock for a single `dial()` before it counts as
    /// one failed try.
    pub dial_timeout: Duration,
    /// Max `dial()` retries on the SAME rung before escalating
    /// (handles transient packet loss).
    pub dial_max_tries: u8,

```

```

    /// Max wall-clock waiting for the WG handshake to complete
    /// after the carrier is up (§5.5).
    pub handshake_timeout: Duration,
    /// Max WG handshake retransmits over an up carrier before
    /// escalating.
    pub max_handshakes: u8,
    /// Hard ceiling on total time spent on ONE rung across all
    /// its tries before forced escalate.
    pub rung_deadline: Duration,
}

impl Default for FailureBudget {
    fn default() -> Self {
        FailureBudget {
            dial_timeout:
                Duration::from_secs(3),    // [04_P0 §4.5 bounded dial;
                tuned in P0]
            dial_max_tries:    2,
            handshake_timeout: Duration::from_secs(5),    // ≈ WG
            handshake retransmit window
            max_handshakes:    3,
            rung_deadline:
                Duration::from_secs(10),    // do not let one rung eat >10s
        }
    }
}

```

UNVERIFIED (calibration): the concrete millisecond values above are **engineering defaults to be calibrated in Phase 0** against the netns rig [04_P0 §8, research-daita_test §5.2–5.3], not values taken from any cited source. Per §11.4.6/§11.4.107(13), thresholds MUST be tuned on the project's own fixtures, not hardcoded from literature. The *structure* (dial vs handshake split, per-rung deadline) is the spec; the numbers are the tunable surface.

4.2 Per-rung cost weighting (smarter than uniform budgets)

plain-udp should fail **fast** (it is either reachable or DPI-blocked within an RTT or two); masque-h3/hysteria2 deserve a slightly longer handshake window because the QUIC handshake itself has more round trips and the GFW residual-censorship window (~500 ms to start blocking, [research- hysteria2 §5(a)]) means a short-lived QUIC handshake can complete inside the uncensored window. The budget is therefore scaled per rung:

```

    /// Multiplier applied to the base budget per rung. UDP fast-fail;
    /// QUIC gets headroom.
    pub fn rung_budget(base: FailureBudget, kind: TransportKind) ->
        FailureBudget {
        let mul = match kind {

```

```

TransportKind::PlainUdp | TransportKind::Lwo
=> 0.7, // fail fast
TransportKind::MasqueH3 | TransportKind::Hysteria2
    | TransportKind::ConnectIp
=> 1.3, // QUIC handshake headroom
TransportKind::Shadowsocks | TransportKind::UdpOverTcp
=> 1.0, // TCP connect+handshake
};
base.scaled(mul) // scales the three Durations; clamps
                integers ≥1
}

```

UNVERIFIED (weights): the 0.7/1.0/1.3 multipliers are design heuristics for Phase-0 calibration, not sourced constants. The *rationale* (UDP fast-fail, QUIC headroom for the ~500 ms residual- censorship window) is grounded in [research-hysteria2 §5(a)].

4.3 Backoff — after the WHOLE order is exhausted

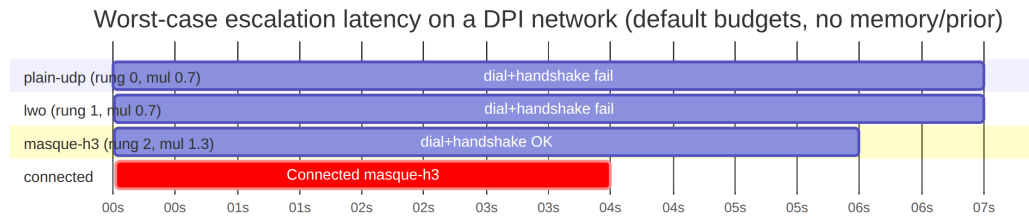
```

#[derive(Clone, Copy, Debug)]
pub struct Backoff {
    pub initial: Duration, // first wait after exhausting the
                          // order
    pub max: Duration,     // ceiling
    pub factor: f32,       // exponential factor
    pub jitter_frac: f32,  // ± fraction of the computed wait
                          // (decorrelates clients, §11.1)
}
impl Default for Backoff {
    fn default() -> Self {
        Backoff { initial: Duration::from_secs(2), max:
            Duration::from_secs(60),
            factor: 2.0, jitter_frac: 0.25 }
    }
}

```

Exhaust-backoff is **per attempt-cycle**, doubling each full-order exhaustion up to max, with $\pm$ jitter_frac so a region-wide outage does not produce a synchronized reconnect stampede on the gateway (§11.1). It resets to initial on any successful Connected.

4.4 Timing budget worked example (default order, hostile network)



Worst case to reach masque-h3 from cold with no memory/prior $\approx$ **rung0 + rung1 deadlines + masque handshake** $\approx 10\text{ s} + 10\text{ s} + \sim 6\text{ s}$ under the default rung_deadline ceiling. **This is exactly the latency per-network memory (§6) and regional priors (§7) exist to eliminate on the second connect** [01-DP §5.3 steps 4–5, 04_P2 §1.4]. After one success the same network reconnects straight to masque-h3 (memory hit), and a CN-resolved client starts at shadowsocks/masque-h3 (prior), never paying the 20 s walk.

5. The driver: dialing, escalation, and the success criterion

5.1 Driver entry points (signatures)

```
// helix-core/src/ladder.rs
pub struct Ladder {
    policy: TransportPolicy,
    state: LadderState,
    memory: Arc<NetworkMemory>, // §6
    status_tx: broadcast::Sender<TunnelStatus>, // [01-DP §5.2]
    telemetry: TelemetrySink, // §9
    next_attempt_id: u64,
}

impl Ladder {
    /// Begin a connect attempt on the current network. Resolves
    /// the seed source (§2.3),
    /// then drives the state machine to `Connected` or `Down`/
    /// `Exhausted-loop`.
    /// `fp` is the current network fingerprint (§6.1). Cancel-
    /// safe: dropping the future
    /// closes the in-flight transport.
    pub async fn connect(&mut self, fp: NetworkFingerprint) ->
        Result<(), LadderError>;
```

```

    /// Replace the active policy (e.g. a regional-prior delta
    arrived, §7). Adopted at the
    /// next attempt boundary if `new.version >
    self.policy.version` (§7.4); never mid-attempt.
    pub fn update_policy(&mut self, new: TransportPolicy);

    /// User pinned/cleared a transport (Mullvad force-mode, §8).
    Triggers an immediate
    /// reconnect using the new pin on the next attempt boundary.
    pub fn set_pin(&mut self, pin: Option<TransportKind>);

    /// Called by the orchestrator when liveness is lost on the
    connected transport (§5.6).
    pub fn on_liveness_lost(&mut self);

    /// Current projected status (also pushed on the broadcast).
    pub fn status(&self) -> TunnelStatus;
}

```

5.2 Mapping TransportConfig from TransportKind + RouteMap

The ladder holds TransportKinds; dial() needs a fully-resolved TransportConfig [01-DP §3.1]. The concrete endpoint, SNI, PSK, and method come from the RouteMap/PeerRoute pushed by the coordinator (doc 03) [01-DP §6.2]:

```

/// Resolve a dialable config for `kind` toward the current
gateway, pulling endpoint/secret
/// material from the live RouteMap. Returns None if the build
lacks the rung's feature or the
/// map lacks required material (→ skip rung, count
`rung_unresolvable`, §9).
fn resolve_config(kind: TransportKind, gw: &GatewayEndpoint, rm:
    &RouteMap)
    -> Option<TransportConfig>;
// e.g. MasqueH3 → TransportConfig::MasqueH3{ url: gw.h3_url, sni:
gw.sni,
//
bind, congestion:
Congestion::Bbr } [01-DP §3.3]
// Shadowsocks → needs gw.ss_psk + method; if absent → None
(skip rung)

```

congestion defaults to Bbr for the mobile/lossy obfuscated rungs (the Hysteria "Brutal" lineage tuning that sustains goodput under loss [01-DP §3.3, research-hysteria2 §1]) and Cubic for the unrestricted baseline.

5.3 The escalation algorithm (pseudocode, normative)

```
fn run_attempt(seed_cursor, attempt_id):
    cursor = seed_cursor
    loop:
        kind = policy.order[cursor]
        cfg = resolve_config(kind, gw, route_map)
        if cfg is None:                                # uncompiled
            feature / missing secret
            telemetry.tick(rung_unresolvable, kind); goto
            escalate_or_exhaust
        budget = rung_budget(policy.budget, kind)
        emit Connecting
        match dial_with_budget(cfg, budget, attempt_id):    # §5.4
            CarrierUp(transport):
                emit Handshaking
                match await_wg_handshake(transport, budget,
                attempt_id):    # §5.5
                    Ok(rtt):
                        memory.remember(fp, kind,
                        policy.version)    # §6.2
                        telemetry.success(kind, escalations =
                        cursor - seed_cursor, region_tag)    # §9
                        emit Connected{ kind, rtt }
                        return Connected(transport)
                    Err(reason):                                #
                        HandshakeFailed/timeout
                        transport.close().await
                        # fall through to escalate
                    Err(reason):                                #
                        DialTimeout/EndpointBlocked
                        # fall through to escalate
                escalate_or_exhaust:
                    if policy.pin.is_some():                    # §8 –
                        pinned, never escalate
                        emit Down{ reason = PinnedTransportFailed(kind) };
                    return Down
                if cursor + 1 < policy.order.len():
                    emit Reconnecting; telemetry.tick(escalated_from,
                    kind)
                    cursor += 1                                # L-I8
                monotonic advance
                continue
            else:
                return Exhausted                                # →
    exhaust_backoff, new_attempt (§4.3)
```

5.4 Cancel-safety & superseded attempts

`dial()` is cancel-safe [01-DP §3.1]; the driver wraps each rung in `tokio::time::timeout` and a `select!` against an `attempt_epoch` watch channel. When `set_pin/update_policy/on_liveness_lost` mints a new `attempt_id`, the epoch bumps, every in-flight `dial()/await_wg_handshake` future is dropped (closing its half-open transport), and any straggler result tagged with a stale `attempt_id` is **discarded** — preventing a late success on an abandoned rung from clobbering a newer attempt (a §11.4.84-class residue at the runtime layer). This is the analogue of Mullvad's "client now randomly selects one of several in-addresses per connection attempt" robustness [research-masque §2], applied to per-rung attempts.

5.5 The success criterion is the WG handshake, NOT the carrier (I2)

A connected Transport only means *bytes can flow*; the rung is "working" **only** when a WireGuard handshake completes over it (helix-wg emits its first decrypted/keepalive verdict) [01-DP §4, I2]. This is load-bearing for DPI realism: a DPI box may let the QUIC/TLS *carrier* connect (so it can fingerprint inside) yet drop the WG datagrams — a carrier-only success would be a PASS-bluff at the ladder layer [§11.4.107 liveness; research-masque §5(d) active probing]. The handshake-complete signal crosses from helix-wg to the ladder via a oneshot the orchestrator wires:

```
/// Resolves when `helix-wg` reports the WG handshake completed
/// over `t`, or errors on timeout.
async fn await_wg_handshake(t: &dyn Transport, b: FailureBudget,
    epoch: u64)
    -> Result<u32 /*rtt_ms*/, EscalationReason>;
```

`rtt_ms` reported on success is the WG-handshake RTT (from `Transport::health().rtt_ewma_ms` [01-DP §3.1]), surfaced in `Connected{rtt_ms}`.

5.6 Liveness loss on a Connected tunnel (downgrade trigger)

While Connected, the orchestrator samples `Transport::health()` [01-DP §3.1]. If `last_rcv_age_ms > liveness_grace` (default 15 s; $\geq$ WG persistent-keepalive interval so a quiet but alive tunnel is not falsely downgraded), the ladder transitions `Connected` $\rightarrow$ `Reconnecting` and re-runs from the seed source. **Fail-static**: the *previous* transport keeps forwarding any traffic WG still encrypts until the new rung is Connected, honoring I3 (control/liveness churn never black-holes an existing tunnel) [01-DP I3].

liveness_grace is calibrated against tc netem loss profiles in Phase 0 so transient loss does not trigger needless re-ladders [research-daita_test §5.2].

6. Per-network memory — remember what worked per network

"On success, **remember the working transport per-network** (SSID / gateway fingerprint) so reconnects on the same hostile network skip straight to what worked — the difference between 'VPN that sometimes connects' and 'VPN that just works'" [01-DP §5.3 step 4, 04_P2 §1.4].

6.1 The network fingerprint (privacy-preserving key)

The memory key **MUST NOT** store the SSID plaintext, BSSID, or gateway endpoint (I5, §11.4.10) — those are identifying. The key is a **keyed hash** of stable network identifiers, so the on-disk cache reveals nothing about *which* networks the user joins:

```
// helix-core/src/netmem.rs
/// Opaque, non-reversible network identity. 16 bytes =
///     BLAKE2s-128 over the network signals,
/// keyed by a device-local secret so two devices on the same SSID
///     produce different keys
/// (defeats cross-device correlation of the cache file). [I5;
///     §11.4.10]
#[derive(Clone, Copy, PartialEq, Eq, Hash)]
pub struct NetworkFingerprint([u8; 16]);

pub struct NetworkSignals<'a> {
    pub ssid: Option<&'a str>, // wifi SSID if
        available (hashed, never stored raw)
    pub gateway_mac: Option<[u8; 6]>, // L2 default-gw MAC
        (best stable LAN identifier)
    pub local_subnet: Option<IpNet>, // e.g. 192.168.1.0/24
    pub connection_type: ConnType, // Wifi | Ethernet |
        Cellular | Unknown
}

impl NetworkFingerprint {
    /// fp = BLAKE2s-128( device_secret || ssid? || gateway_mac?
    ///     || subnet? || conntype )
    /// Cellular networks (no stable LAN id) fold the carrier/
    ///     conntype only → coarse but stable.
    pub fn derive(sig: &NetworkSignals, device_secret: &[u8; 32])
        -> NetworkFingerprint;
}
```

```
#[derive(Clone, Copy, PartialEq, Eq)] pub enum ConnType { Wifi,
    Ethernet, Cellular, Unknown }
```

UNVERIFIED (algorithm choice): BLAKE2s-128 + device-keying is a spec design choice consistent with WireGuard's own BLAKE2s usage [01-DP §4] and the I5/§11.4.10 privacy constraint; no cited source prescribes the exact fingerprint construction. Mullvad's internal per-network-memory representation is not public — treat this layout as HelixVPN's own design, not a parity reproduction.

6.2 The memory store (signatures)

```
// helix-core/src/netmem.rs
pub struct NetworkMemory { /* in-mem LRU + atomic-write
    persistence (§6.4) */ }

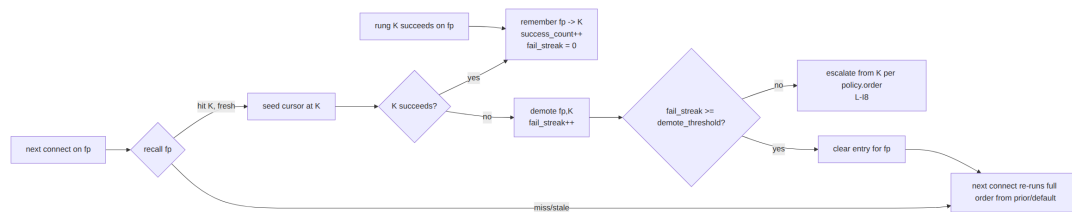
impl NetworkMemory {
    /// Record that `kind` succeeded on `fp` under policy
    /// `policy_version`.
    /// Idempotent; updates `last_good` + bumps `success_count`,
    /// refreshes `last_seen_epoch_day`.
    pub fn remember(&self, fp: NetworkFingerprint, kind:
        TransportKind, policy_version: u64);

    /// Look up the remembered starting rung for `fp`, if any AND
    /// not stale (§6.5).
    /// Returns the rung to seed the cursor at (§2.3 step 2). None
    /// => fall through to prior/default.
    pub fn recall(&self, fp: NetworkFingerprint, now_epoch_day:
        u32) -> Option<TransportKind>;

    /// On repeated failure of the remembered rung, demote it so
    /// next time we don't re-pay it
    /// (§6.3). After `demote_threshold` consecutive failures the
    /// entry is cleared.
    pub fn demote(&self, fp: NetworkFingerprint, failed:
        TransportKind);

    /// Persist the in-mem map to disk atomically (write-
    /// temp+rename). Called debounced (§6.4).
    pub fn flush(&self) -> std::io::Result<()>;
}
```

6.3 Remember/demote lifecycle



Crucially, after a memory hit the ladder still **escalates upward from the remembered rung** if it fails (it does not reset to order[0]) — the remembered rung was the *minimum* that worked, so a network that got worse only needs to climb further, not restart from the bottom [01-DP §5.3 step 4].

6.4 On-disk byte format (the persisted cache)

The cache is a small, fixed-layout binary file at the platform config dir (\$XDG_CONFIG_HOME/helixvpn/netmem.bin on Linux; analogous per-platform), chmod 600, listed in .gitignore with recall() itself as the §11.4.77 "regenerate by re-learning" mechanism (it is a pure cache — losing it costs one slow connect, never correctness):

netmem.bin (little-endian, append-tolerant, atomically rewritten via temp+rename)

Header (16 bytes)			
magic	u32	= 0x484C4E4D	
("HLNM")			
format_ver	u16	=	
1			
reserved	u16	=	
0			
entry_count	u32		
crc32	u32	(over all entries; mismatch ⇒ discard whole file)	
Entry[entry_count] (24 bytes each)			
fingerprint	[u8;16]	BLAKE2s-128 network key	
(\$6.1)			
last_good	u8	TransportKind::code()	
(\$2.1)			
fail_streak	u8	consecutive failures of last_good	
success_cnt	u16	capped at	
u16::MAX			
last_seen	u32	days since unix epoch (coarse → no fine	

timing) |

No plaintext SSID, no endpoint, no per-connection

timestamp — only the irreversible fingerprint, a 1-byte rung, and a day-granularity recency (coarse enough not to be a behavioral log, satisfying I5 [01-DP I5, §11.4.10]). Total size for 256 remembered networks $\approx 16 + 256 \times 24 \approx 6.1$ KiB. Writes are debounced (≤ 1 flush / 5 s) and atomic (temp file + rename) so a crash mid-write never corrupts the cache (the CRC + magic gate discards a torn file and re-learns).

6.5 Staleness, eviction, and the network-class fast-skip

- **Staleness:** `recall()` ignores entries older than `memory_ttl_days` (default 30) — a transport that worked a month ago on a network whose DPI policy has since changed is weak evidence; re-learn.
 - **Eviction:** LRU by `last_seen`, capped at `max_entries` (default 256) — bounded disk + bounded cross-network correlation surface.
 - **Network-class fast-skip:** if the current network has *already* demonstrated a hard UDP block (any UDP/QUIC rung failed with `EndpointBlocked` this session), the ladder may skip the remaining `needs_udp() == true` rungs and jump to `shadowsocks/udp-over-tcp` — because "QUIC/Hysteria2/ MASQUE-over-H3/ WireGuard all fail [on a hard UDP block]" [research-hysteria2 §5(b)]. This is a *within-attempt* optimization (still monotonic, L-I8): once UDP is proven dead, climbing through three more UDP rungs is wasted latency.
-

7. Coordinator-pushed regional priors

"The coordinator may push a **regional prior** (e.g. clients from CN-resolved IPs start at `shadowsocks`) via `TransportPolicy.order`, so users in censored regions do not pay the escalation latency every time" [01-DP §5.3 step 5, 04_P2 §1.4].

7.1 Source: a transport_prior field on the NetworkMap (doc 03 owns the protobuf)

The regional prior rides the **existing** WatchNetworkMap server-stream [01-DP §6.2, SYNTHESIS §2] — no new channel (push, don't poll; reconcile, don't restart [01-DP §6.3, 04_ARCH §4.4]). This document specifies only the **shape the ladder consumes**; doc 03 owns the wire protobuf:

```
// helix-route/src/map.rs – additive field the reconciler surfaces
// to the ladder
pub struct RouteMap {
    // ... existing fields [01-DP §6.2] ...
    pub transport_prior: Option<TransportPrior>, // pushed
        regional prior; None ⇒ use default
}

/// The coordinator's recommended ladder for THIS client on THIS
/// network leg.
pub struct TransportPrior {
    pub order: Vec<TransportKind>, // recommended escalation
        order (e.g. [shadowsocks, masque-h3, udp-over-tcp] for CN)
    pub version: u64, // monotonic; ladder
        ignores ≤ current (§7.4)
    pub leg: LegId, // user↔gateway vs
        gateway↔connector (per-leg, D6 §11.3 of doc 01)
    pub reason: PriorReason, // why (for UX +
        telemetry; NOT per-user data)
}
pub enum LegId { UserToGateway, GatewayToConnector }
pub enum PriorReason { GeoCensored, OperatorPolicy,
    DefaultUnrestricted, LearnedAggregate }
```

The coordinator derives the prior **server-side** from the client's *resolved source region* (it already terminates the connection, so it sees the source IP's geo without the client reporting it) and from the **aggregate** censorship-evasion telemetry (§9) — "transport X succeeded after N escalations in region R" feeds back into the prior for region R, closing the loop [01-DP §5.3 step 6, 04_ARCH §9]. No per-user history is consulted; the prior is a region-wide statistic.

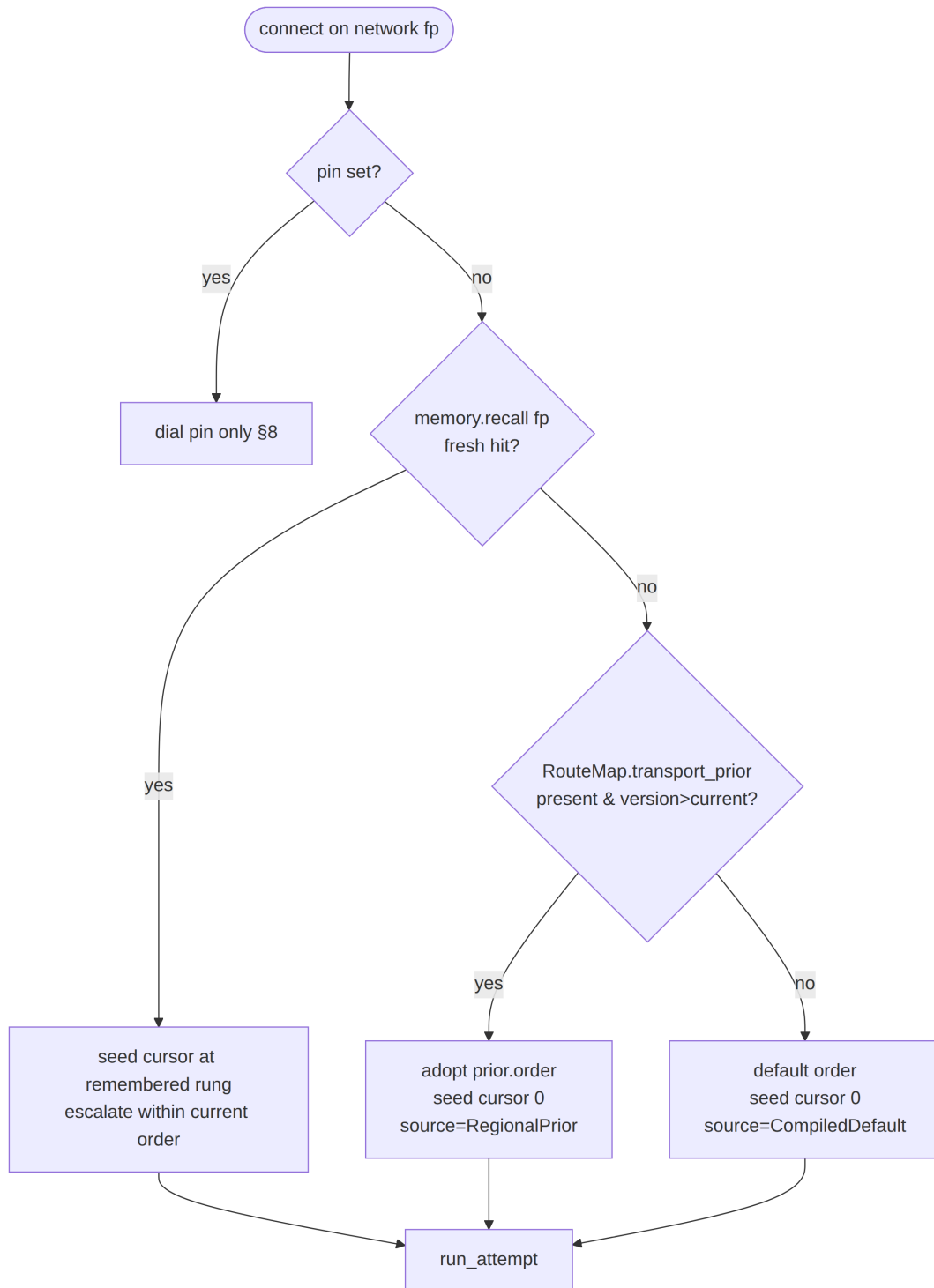
7.2 Worked regional-prior table (illustrative, coordinator-configurable)

Resolved region / condition	Pushed order	Rationale (sourced)
Unrestricted	[plain-udp, lwo, masque-h3, ...] (= default)	fast path first [01-DP §5.3]

Resolved region / condition	Pushed order	Rationale (sourced)
CN (GFW QUIC-SNI block)	[masque-h3, shadowsocks, udp-over-tcp] or [hysteria2, ...]	GFW blocks QUIC by SNI but MASQUE survives iff the H3 proxy SNI is unblocked; Salamander/Gecko-obfuscated QUIC survives by hiding SNI [research-hysteria2 §5(a), §6]
Hard-UDP-block region/corp	[shadowsocks, udp-over-tcp]	"QUIC/.../WireGuard all fail ... need UDP" [research-hysteria2 §5(b)]
Operator override	exactly as configured	OperatorPolicy

UNVERIFIED (specific orders): the *exact* per-region orders are coordinator policy data, tuned by operators against live conditions and the §9 aggregate feedback — they are **not** fixed by this spec. The table illustrates the mapping logic grounded in the cited censorship-regime research; the values are examples, not normative constants.

7.3 Prior vs memory vs default — combined decision



Note the **order** that a memory hit escalates within is still the *currently adopted* policy order (prior-or-default) — memory chooses the *start rung*, the prior/default chooses the *escalation sequence above it* (§6.3). This composes cleanly: a CN client

whose memory says "shadowsocks worked" seeds at shadowsocks but, if that fails, climbs udp-over-tcp per the CN prior, never wandering back to plain-udp.

7.4 Reconcile semantics (push-don't-poll, no restart)

The ladder adopts a new prior via `update_policy()` (§5.1) **only** when `new.version > self.policy.version` (monotonic; stale/replayed deltas ignored — the §11.4.116 "snapshot reporting a lower version is a contradiction" discipline). Adoption takes effect at the **next attempt boundary** (L-I8), never mid-attempt; a prior arriving while Connected does **not** tear down a working tunnel — it is staged and applied on the next Reconnecting/manual reconnect. Convergence target inherited from the control plane: a pushed prior reflected on the client in **< 1 second** of the delta arriving [04 ARCH §4.4, 01-DP §6.3].

8. Manual override (pin) — Mullvad force-mode parity

Mullvad lets the user **force** a transport ("mullvad obfuscation set mode quic") [research-masque §2, research-hysteria2 §2]; HelixVPN mirrors this with TransportPolicy.pin [01-DP §5.3]:

- `set_pin(Some(kind))` → the ladder dials **only** kind, with the rung budget, and on failure goes straight to `Down{ PinnedTransportFailed(kind) }` — **no auto-escalation** (the user asked for exactly this transport; silently using a different one would violate their explicit intent and could be a privacy regression, e.g. falling back from an obfuscated rung onto plain-udp on a monitored network). The UX surfaces the failure and the option to clear the pin.
- `set_pin(None)` → return to auto (ladder + memory + prior).
- A pin is honored on **every** reconnect until cleared (it is policy, not per-attempt state).
- pin does **not** write per-network memory on success (a forced choice is not evidence about what the network *requires*, only about what the user *preferred*) — so clearing the pin later falls back to genuine learned/prior behavior, not a memory entry biased by the override. (Telemetry still counts a pinned success under `source=OperatorOverride`, §9.)

```
#[derive(Clone, Debug)]
pub enum DownReason {
    PinnedTransportFailed(TransportKind), // §8 – no escalation
        by design
    AllRungsExhausted, // every rung failed;
        in exhaust-backoff loop
}
```



```

        NoRouteMap,                                // map not yet
        received (control plane bootstrap)
KillSwitchEngaged,                                // fail-closed: egress
        blocked, tunnel intentionally down
NonRetryable(String),                             // config/build error
        (e.g. uncompiled pinned feature)
}

```

9. Telemetry — aggregate only, no per-user data (I5)

"Telemetry records **only** 'transport X succeeded after N escalations in region R' (aggregate, no per-user data, I5) → feeds the Censorship-Evasion Success dashboard" [01-DP §5.3 step 6, 04_ARCH §9].

```

// helix-core/src/ladder_telemetry.rs
/// A single aggregate-safe outcome. Emitted at most once per
/// connect attempt. Contains NO
/// fingerprint, NO endpoint, NO timestamp finer than the day, NO
/// SSID — only counters + coarse tags.
#[derive(Clone, Debug)]
pub struct LadderOutcome {
    pub winning_kind:
        Option<TransportKind>, // None ⇒ exhausted (no rung
                               worked)
    pub escalations: u8,        // rungs climbed
                               before success (cursor - seed)
    pub region_tag: RegionTag,  // coarse, k-
                               anonymized bucket (e.g. "CN", "EU", "XX")
    pub policy_source: PolicySource, // default /
                               regional-prior / operator-override
    pub udp_blocked: bool,      // did any
                               needs_udp rung hit EndpointBlocked?
}

pub trait TelemetrySink: Send + Sync {
    fn record(&self, o: LadderOutcome); // batched,
    // sampled, sent to control plane
    fn tick(&self, counter: &'static str, kind:
        TransportKind); // e.g. "rung_unresolvable"
}

```

Hard constraints (CI-enforced per [01-DP I5, SYNTHESIS §7] — "CI schema-lint fails build if any durable connection/traffic/packet table appears"):

- `region_tag` is a **k-anonymized coarse bucket** (country/region class), never a city/ASN/IP; a bucket with < k clients in the window is reported as XX (no-region). UNVERIFIED: the concrete k is an operator privacy parameter, not a sourced constant.

- No LadderOutcome field may be joined to a device id, user id, or fingerprint at the sink.
- The escalation **event trace** (TunnelStatus sequence) is captured **locally** as anti-bluff evidence (§10.4) and is NOT shipped — only the post-hoc aggregate LadderOutcome is.

10. Edge cases, security, performance, and test points

10.1 Error → ladder-action taxonomy

The doc-01 TransportError [01-DP §3.1, error.rs] maps to ladder actions as follows:

TransportError	Phase	Ladder action
DialTimeout	dial	retry on same rung up to dial_max_tries; then escalate
HandshakeFailed(_)	dial/WG	escalate (carrier or WG handshake refused)
EndpointBlocked	dial	escalate and set session udp_blocked if needs_udp() (enables §6.5 fast-skip)
Oversize(_)	runtime	NOT an escalation trigger — lower inner WG MTU [01-DP §10]; ladder stays put
Closed	runtime (connected)	treat as liveness-lost → Reconnecting (§5.6)
Io(_)	any	transient: retry within dial_max_tries; persistent → escalate
Quic(_)	dial/runtime (QUIC rungs)	escalate (QUIC handshake/transport error)

10.2 Enumerated edge cases (each is a test point, §10.4)

1. **Carrier connects but WG never handshakes** (DPI lets QUIC through to fingerprint, drops WG) → Handshaking times out → escalate; NOT a false success (§5.5) [research-masque §5(d)].
2. **Remembered rung now blocked** → demote, climb from it, clear after demote_threshold (§6.3).

3. **Prior references an uncompiled rung** (hysteria2 without the feature) → filter it, count `policy_rung_unavailable`, continue with the subset (§2.4).
4. **Hard UDP block mid-order** → after first `EndpointBlocked` on a UDP rung, fast-skip remaining UDP rungs to the TCP carriers (§6.5) [research-hysteria2 §5(b)].
5. **Policy delta arrives while Connected** → staged, applied at next attempt boundary, tunnel not torn down (§7.4, L-I8).
6. **Network changes mid-session** (roam Wi-Fi → cellular) → fingerprint changes → orchestrator triggers `on_liveness_lost` → re-ladder under the new fp's memory/class (§6.1, WG roaming latches the new path [01-DP §8.2]).
7. **All rungs exhausted** → exhaust-backoff with jitter, restart from seed (§4.3); never a tight spin loop (host-safety §12, §11.1).
8. **Pinned-feature missing at runtime** → `Down{ NonRetryable }`, not a silent fallback (§8).
9. **Torn netmem.bin** → CRC/magic mismatch discards file, re-learns; never a crash, never corrupt recall (§6.4).
10. **Stale superseded dial() result** → discarded by `attempt_id` epoch (§5.4).

10.3 Security considerations

- **No privacy regression on override/escalation.** The ladder MUST NOT silently downgrade from an obfuscated rung to a less-evasive one *while a pin is set* (§8); in auto mode, escalation only ever moves toward *more* evasive rungs (monotonic, L-I8) — it never falls *back* to plain-udp within an attempt once an obfuscated rung is reached.
- **Fingerprint cache is non-identifying.** Device-keyed BLAKE2s-128, no SSID/endpoint/fine-timing (§6.1, §6.4) — the cache file leaks nothing about which networks the user joins, even if exfiltrated (§11.4.10).
- **Telemetry cannot deanonymize.** Coarse k-anonymized region bucket, no device join key, schema-lint forbids durable per-connection rows (§9, [SYNTHESIS §7]).
- **DPI-realism is a probe, not an assumption.** The success criterion is the WG handshake over the carrier (§5.5); the wire-fingerprint of the *carrier* (e.g. `masque-h3` classifies as HTTP/3 with no WG signature via `tshark`) is verified by the SEC test, not assumed [01-DP §12.1 G2, research-masque §5].
- **GFW residual-censorship awareness.** Budgets give QUIC rungs handshake headroom because the GFW's QUIC block is *residual* (≈ 500 ms to start, 3-min block) so a fast QUIC handshake can complete in the uncensored window [research-hysteria2 §5(a)]; the ladder must not abandon a QUIC rung before that window has had a fair chance (§4.2).
- **No-guessing on cause.** A rung failure is classified from the concrete `TransportError` (§10.1), never reported as "probably

blocked" — EndpointBlocked vs DialTimeout vs HandshakeFailed are distinct, captured facts (§11.4.6).

10.4 Performance budget

Metric	Target	Source / rationale
Memory-hit reconnect (warm)	seed straight to last-good rung; 0 wasted escalations	the entire point of §6 [04_P2 §1.4]
Prior-seeded first connect (censored region)	start at the region's working rung; 0-1 escalations typical	§7 [01-DP §5.3 step 5]
Cold worst-case (no memory/prior, DPI net)	bounded by Σ rung_deadline to the working rung (≈ 20 s to masque-h3, §4.4)	§4.4
Ladder CPU overhead	negligible vs transport crypto; the ladder is control logic, not a data path	the data path is helix-wg+transport [01-DP §3,§4]
netmem.bin size	$\leq \sim 6$ KiB for 256 networks; ≤ 1 flush/5 s debounced	§6.4
Exhaust-backoff	exponential 2 s $\rightarrow$ 60 s $\pm 25\%$ jitter; no tight spin	§4.3, §11.1 host-safety

The MASQUE throughput tax itself ("computationally very expensive ... affects throughput" [research- masque §2]) is a **transport** budget owned by doc 01 §10/§11; the ladder only *chooses* the rung, so its job is to spend the cheapest rung that works — which is precisely why the default order is cheapest-first and memory/prior exist to avoid over-escalating.

10.5 Test points (tied to §11.4.169 closed test-type vocabulary)

Using the §11.4.169 code map [06-P0 §0.4]: UT unit, IT integration, E2E, FA full-automation/deterministic, SEC security, SC stress+chaos, CONC concurrency, RACE race/deadlock, BENCH performance, CH Challenges, HQA HelixQA. Every PASS carries captured evidence per §11.4.5/.69/.107 — for the ladder the evidence is the **ordered TunnelStatus event trace** [01-DP §12.1].

#	Test	Type(s)	Evidence captured
T1		UT,CONC	

#	Test	Type(s)	Evidence captured
	Escalation order walked on repeated HandshakeFailed (mock transports)		ordered TunnelStatus trace Connecting→Reconnecting...→Connected{masque-h3}
T2	Carrier-up-but-WG-never-handshakes → escalate, not false success (§5.5)	UT,SEC	trace shows Handshaking→Reconnecting, never Connected on the dead rung
T3	Per-network memory: 2nd connect on same fp seeds at last-good rung, 0 escalations (§6)	IT,E2E,FA	two traces; 2nd has escalations=0; netns rig [research-daita_test §5.1]
T4	Regional prior pushed → CN-tagged client starts at prior order[0] (§7)	IT,E2E	RouteMap delta + trace showing seed at prior rung
T5	Pin forces rung, no escalation on failure (§8)	UT,SEC	trace ends Down{PinnedTransportFailed}, no other rung dialed
T6	Hard-UDP-block fast-skip to TCP carriers (§6.5) over nft drop udp	IT,SEC	nftables UDP-drop + trace skips udp rungs [research-daita_test §5.4]
T7	Deterministic: T1/T3 produce identical traces over N=3 runs (§11.4.50)	FA	three byte-identical evidence hashes
T8	Cancel-safety: pin/policy change mid-dial discards stale result (§5.4)	RACE,CONC	loom/-race; no stale Connected after epoch bump
T9	netmem.bin torn-write recovery + atomic flush (§6.4)	SC	kill mid-flush; next boot recalls cleanly OR discards torn file
T10	Exhaust-backoff bounded, jittered, no spin loop (§4.3)	SC,BENCH	timer trace; CPU sample stays flat during backoff
T11	End-to-end escalation through a real	E2E,CH,HQA	edge+connector booted via containers submodule [01-DP §12.1 G2]

#	Test	Type(s)	Evidence captured
T12	DPI-sim slice to a live edge Paired §1.1 mutation: break <code>await_wg_handshake</code> to accept carrier-only → T2 MUST fail	UT (meta)	mutation makes the gate FAIL (proves the gate is real)

Per §11.4.169 the only permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason (e.g. MEM on a host with no iOS device → `topology_unsupported`), never a silent gap.

11. How this mirrors Mullvad's UX

Mullvad behavior (cited)	HelixVPN ladder mechanism
Auto-tries QUIC obfuscation after a few failed normal connection attempts [research-masque §2, research-hysteria2 §2]	Default order puts <code>masque-h3</code> after <code>plain-udp/lwo</code> ; the failure budget (§4) is the "few failed attempts" before escalation
User can force a transport (mullvad obfuscation set mode <code>quic</code>) [research-masque §2]	<code>TransportPolicy.pin</code> — dials only that rung, no auto-escalation (§8)
Client randomly selects one of several in-addresses per connection attempt for robustness [research-masque §2]	<code>attempt_id</code> epoching + per-attempt re-resolution from <code>RouteMap.endpoint_candidates</code> (§5.4); endpoint selection is doc-01's NAT-traversal field [01-DP §6.2, §8]
Positioned for "restrictive networks that allow web traffic only" [research-masque §2]	Regional prior seeds censored-region clients straight at the web-blending rung (§7); never pays the cold-walk
Throughput cost acknowledged, "use only when needed" [research-masque §2]	Cheapest-rung-first order + per-network memory ensures the expensive obfuscation is used only where it is actually required (§2.1, §6, §10.4)

Mullvad behavior (cited)	HelixVPN ladder mechanism
Server-pushed defense selection (DAITA v2 negotiated model) [research-daita_test §4]	Same control-plane push pattern: the coordinator pushes the prior as NetworkMap data, no client rebuild (§7.1, §7.4)

11.1 Host-session-safety note (constitution §12)

The exhaust-backoff (§4.3) with jitter, the debounced cache flush (§6.4), and the bounded per-rung deadlines (§4.1) jointly guarantee the ladder is a **low, bounded** CPU/IO load even under a total regional outage — no tight reconnect spin, no unbounded disk churn, no memory growth (LRU-capped cache §6.5). This satisfies §12 host-safety and the §12.6 $\leq 60\%$ memory ceiling for the client process.

12. Frozen-contract surface & cross-document seams

Artefact	Owner	This doc's relation
Transport trait, TransportConfig, dial(), TransportError	doc 01 §3 (frozen)	consumed verbatim; never modified
TunnelStatus broadcast enum	doc 01 §5.2 (frozen)	projected onto (§3.2)
RouteMap / PeerRoute / endpoint_candidates	doc 01 §6.2 (frozen)	source of resolved configs (§5.2) + regional prior (§7.1)
WatchNetworkMap protobuf delivering transport_prior	doc 03	this doc specifies only the <i>shape consumed</i> ; doc 03 owns the wire
Per-leg TransportPolicy (D6 asymmetric topology)	doc 01 §11.3	LegId on the prior (§7.1) makes per-leg ladders free
FFI re-emission of TunnelStatus + pin command	doc 05	out of scope; surfaced via §3.2 / §5.1
§11.4.169 test-type vocabulary	doc 06/07/10	§10.5 cites the closed-set codes

New types introduced **by this document** (additive to doc 01, not modifications): TransportKind, TransportPolicy (refined), FailureBudget, Backoff, LadderState, EscalationReason, DownReason,

NetworkFingerprint, NetworkSignals, NetworkMemory, TransportPrior, LegId, PriorReason, LadderOutcome, RegionTag, PolicySource, TelemetrySink, and the Ladder driver. All live in helix-core/src/ladder.rs, helix-core/src/netmem.rs, and helix-core/src/ladder_telemetry.rs [01-DP §2, §12]; the transport_prior field is additive on helix-route/src/map.rs.

End of nano-detail specification — Automatic Transport Selection Ladder (Volume 2, Data Plane). Deepens [01-DP §5.3, §5.4]. Pair with doc 03 (WatchNetworkMap — the live source of the transport_prior seam §7.1) and doc 01 (the frozen Transport/TunnelStatus/RouteMap contracts). Every UNVERIFIED mark flags a value/structure that is HelixVPN design rather than a sourced fact, per constitution §11.4.6 — those are the calibration surfaces Phase 0 resolves with captured evidence, not guesses to ship.